POOJA ANBALAGAN

Data Science Undergraduate
| AI/ML Engineer | Data Scientist | MLOps DevOps | Cloud Engineer |
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SUMMARY

Data Science undergraduate at Sabaragamuwa University of Sri Lanka with hands-on experience in machine learning, model evaluation, and deployment using Python, Docker, Kubernetes, and GitHub Actions. Developed a sentiment analytics platform (0.8906 macro-F1) and a crop disease classifier (95.28% Top-1 accuracy). First-prize winner at IEEE WIE AURELIYA 2.0 and multiple-time hackathon finalist. Seeking internship opportunities in AI/ML Engineering, Data Science, MLOps, DevOps, or Cloud Engineering.

EDUCATION

B.Sc. (Hons) in Data Science, Sabaragamuwa University of Sri Lanka *2024 – 2028*
Current GPA: 3.36
Relevant coursework: Statistics, Machine Learning, Artificial Intelligence, Databases, Cloud Computing, Data Mining, Business Intelligence, Software Engineering, Networking

TECHNICAL SKILLS

- **Programming Languages:** Python, Java, SQL, JavaScript
- **Machine Learning & Artificial Intelligence:** Scikit-learn, PyTorch, TensorFlow, Supervised & Unsupervised Learning, Classification, Clustering, Natural Language Processing (NLP), Transformers (DistilBERT), Transfer Learning, Model Optimization & Evaluation
- **Data Science & Analytics:** Pandas, NumPy, Matplotlib, Seaborn, Plotly, Power BI, Exploratory Data Analysis (EDA), Data Cleaning, Feature Engineering, Statistical Analysis
- **MLOps & DevOps:** Docker, Docker Compose, Kubernetes, Git, GitHub Actions, CI/CD Pipelines, Model Deployment, ONNX, Linux
- **Frontend Development:** React.js, HTML5, CSS3, Responsive Web Development
- **Backend Development & API Engineering:** FastAPI, Django REST Framework, RESTful API Development
- **Databases & Data Management:** PostgreSQL, MySQL, SQLite
- **Cloud Platforms & Development Tools:** AWS, Vercel, Netlify, Jupyter Notebook, Google Colab, VS Code, GitHub

PROJECTS

Customer Review Sentiment Analytics Platform | Full-Stack ML Platform *Aug 2026*
GitHub: github.com/PoojaAnbalagan/SENTIMENTAL-ANALYSIS
Live: Customer Review Sentiment Analytics Platform
Tech*: *Python, FastAPI, Scikit-learn, DistilBERT, React, Docker, Kubernetes, GitHub Actions

- Built a full-stack NLP platform using FastAPI and React to deliver real-time customer review sentiment predictions and analytics.
- Benchmarked traditional ML models, VADER, and DistilBERT, achieving **90.93% accuracy and 0.8906 Macro-F1** with DistilBERT.
- Containerized services using Docker and Docker Compose, with Kubernetes deployment manifests.
- Implemented REST APIs, health-monitoring endpoints, and GitHub Actions CI workflows.

AgroGuard-AI | Transfer Learning & Crop Disease Detection *Sep 2026 | Team project, CodeArena'26*
GitHub: github.com/SarasiRajarathna/AgroGuard-AI
***Tech:** Python, PyTorch, MobileNetV3-Small, ONNX, Node.js, CLI inference*

- Developed a 112-class crop disease classification model using MobileNetV3-Small transfer learning.
- Built a reproducible ML pipeline with data validation, class-imbalance handling, augmentation, and leakage prevention.
- Achieved **95.28% Top-1 Accuracy and 94.23% Macro-F1** on the held-out test set.
- Exported the model to ONNX, achieving approximately **18.5 ms/sample CPU inference latency**, with CLI and Node.js integration.

CrowdVibe | Real-Time AI Crowd-Safety Platform *Jul 2026 | Team project, HackElite 3.0*
GitHub: github.com/aradhya534/CrowdVibe | Live: crowd-vibe-seven.vercel.app/
***Tech:** Python, FastAPI, Scikit-learn (DBSCAN), WebSockets, React, Vite, relational database, Vercel*

- Developed a real-time crowd-safety platform using FastAPI, React, and WebSockets.
- Applied DBSCAN spatial clustering to identify crowd-density risks from positional data.
- Implemented an asynchronous simulation engine processing 500+ position updates per second, with automated safety alerts and route recommendations.
- Contributed to technical documentation and project communication.

Customer Churn Prediction on AWS | Cloud-Based ML Data Pipeline *Oct 2026*
***GitHub:** github.com/PoojaAnbalagan/ChurnPredictionAWS*
***Tech:** AWS S3, AWS Lambda, Amazon CloudWatch, Python, Boto3, Pandas*

- Designed an event-driven data pipeline using Amazon S3 and AWS Lambda for automated prediction-output validation.
- Configured S3 event triggers, IAM permissions, and CloudWatch logging for secure execution and monitoring.
- Organized raw, processed, and prediction datasets into separate S3 storage prefixes.

ACHIEVEMENTS & ACTIVITIES

- **1st Prize - IEEE WIE AURELIYA 2.0 Innovation Competition** (https://lnkd.in/p/gqM2Enxp)
- **2nd Place - CodeRally 7.0 (24-Hour Hackathon)** (https://lnkd.in/p/gXHb-W6h)
- **CodeArena'26 – Finalist** (https://lnkd.in/p/gv5YFKf6)
- **CryptX Hackathon – Finalist** (https://lnkd.in/p/gqG3yWxu)
- **J'puraXtreme 3.0 (24-Hour Hackathon) - Finalist** (https://lnkd.in/p/gNshHGaT)
- **HackElite 3.0 - Phase 2 Shortlist** (https://lnkd.in/p/g9btcpsi)

LEADERSHIP & VOLUNTEERING

- **Chairperson, VisionVibe Magazine, Rotaract Club of SUSL (2026/2027)**.
- **Co-Chairperson and Event Coordinator, Rotaract Club of SUSL**(https://lnkd.in/p/ghTGRv3t)
- **Announcer:** hosted Rotaract Club events and programs, as well as IEEE and Computer Society events. (https://lnkd.in/p/ghA8n-V9)
- **Content Writer and Graphic Designer, Rotaract Club of SUSL**(https://lnkd.in/p/g88DaCEf)

REFERENCES

Professor (Dr.) S. Vasanthapriyan, Ph.D, SMIEEE - Professor in Computer Science
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